Probing and prompting LLM's

Report - Assignment 3 part 2

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Methodology

In this part of the assignment, I used a Large Language Model through the Ollama Python library to classify song lyrics into their associated genre. I implemented two prompting strategies: zero-shot prompting and few-shot prompting. The model used in this experiment was llama3.2:1b, which was selected due to hardware and memory limitations.
To reduce runtime, I made several adjustments. First, instead of using the full lyrics, I truncated the lyrics to the first 200 characters. This was necessary because longer inputs significantly increased processing time and sometimes caused unstable responses from the smaller model. Second, I initially selected three examples per genre for the few-shot prompting strategy. However, running the model with three examples caused extremely long runtimes (more than six hours). Therefore, I reduced this to two examples per genre. I also implemented runtime measurements to ensure that the code would complete execution within time constraints.
For testing, I first used a small sample of the dataset to verify that the prompting and evaluation pipeline worked correctly before scaling up the experiment.
Before calculating Precision, Recall, and F1 score, I inspected the raw outputs of the model. This was necessary because the model did not always return clean genre labels but sometimes produced explanations or slightly different label formats. Therefore, I normalized the outputs to ensure that they matched the dataset labels before evaluation. The performance of both prompting strategies was evaluated using Precision, Recall, and F1 score, as well as confusion matrices to investigate whether performance varied across genres.

Results

The performance of the model was evaluated using Precision, Recall and F1 score for both zero-shot and few-shot prompting. Overall performance was low for both strategies, but few-shot prompting showed a small improvement compared to zero-shot prompting. For zero-shot prompting, the overall accuracy was *0.06,* with a macro F1 score of *0.04.* The model showed very uneven performance across genres. The highest recall was observed for Country (0.95), but this was mainly because the model predicted Country very frequently, resulting in very low precision (0.06). Most other genres such as Rock, Pop, Metal and Jazz had near-zero recall and F1 scores, indicating that the model rarely predicted these classes correctly.
Few-shot prompting improved performance slightly. Accuracy increased to *0.10* and the macro F1 score increased to *0.07.* Some genres such as Pop (F1 = 0.14), Rock (F1 = 0.10), and Country (F1 = 0.13) showed modest improvements compared to zero-shot prompting. However, several genres such as Indie, Jazz and R&B still had F1 scores of zero, indicating that the model struggled to distinguish these genres even with examples.
The confusion matrices further show that the model frequently confused genres. In the zero-shot setting, many songs from different genres were predicted as Country, suggesting that the model defaulted to this category when uncertain. In the few-shot setting, predictions were more distributed across genres, but confusion between similar genres such as Rock, Pop and Metal remained common.

Discussion

The results show that few-shot prompting performed better than zero-shot prompting, although the improvement was quite small. This slight improvement is likely because the few-shot examples helped the model better understand the expected output format and the relationship between lyrics and genres. However, because only two examples per genre were used due to runtime limitations, the model may not have received enough information to generalize well.
The model struggled particularly with genres that are probably linguistically similar, such as Rock, Pop and Metal. In contrast, Country showed relatively high recall, possibly because country lyrics often contain distinctive thematic elements such as rural life or storytelling structures. However, the very low precision indicates that the model overpredicted this genre, which could be because it has most examples.
Another important factor affecting performance is the size of the model. The llama3.2:1b model was chosen due to hardware limitations, but smaller models generally have weaker classification capabilities than larger models. This likely contributed to the overall low F1 scores. Additionally, only the first 200 characters of each lyric were used to reduce runtime. While this made the experiment feasible, it may also have removed important contextual information needed for correct classification.

Performance could likely be improved in several ways. Using a larger model would likely improve classification accuracy. Increasing the number of few-shot examples could also help the model better understand genre distinctions, although this would increase runtime. Using longer lyric excerpts could provide more contextual information. Finally, prompt engineering improvements such as genre definitions could potentially improve results.
Overall, the experiment shows that few-shot prompting can improve genre classification performance compared to zero-shot prompting, but that performance remains limited when using a small model and short text inputs. The results also demonstrate that genre classification difficulty varies across genres.